

José Ferreira

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EDUCATION

PhD in Physics <i>University of Aveiro</i>	09/2023 - PRESENT
MSc in Physics <i>Faculty of Sciences of the University of Lisbon</i>	09/2020 - 09/2022
BSc in Physics <i>Faculty of Sciences of the University of Lisbon</i>	09/2017 - 07/2020

RESEARCH INSTITUTES

Gr@v Gravitation group at the <i>Center for R&D in Mathematics and Applications</i> of the University of Aveiro, Portugal.	09/2023 - PRESENT
Institute of Astrophysics and Space Sciences Cosmology group at the <i>Institute of Astrophysics and Space Sciences</i> , Lisbon node, Portugal.	09/2021 - 07/2023

TEACHING

Unviersity of Aveiro Courses: Numerical and Statistical Methods	09/2025 - PRESENT
Faculty of Sciences of the University of Lisbon Courses: Experimental Physics I, Elements of Physics, Mathematical Methods for Physicists, General Physics	02/2022 - 07/2023

PUBLICATIONS

- J. Ferreira, G. Bozzola, C. Herdeiro, V. Paschalidis, M. Zilhão. “Electromagnetic duality degeneracy in dynamical black hole mergers”. (19 May 2026). [arxiv:2605.20493](https://arxiv.org/abs/2605.20493)
- J. Ferreira, C. Herdeiro, E. Radu, M. Zilhão. “Spontaneous spherical symmetry breaking of black holes with resonant hair”. (24 April 2026). [arxiv:2604.22437](https://arxiv.org/abs/2604.22437)
- J. Nicoules, J. Ferreira, C. Herdeiro, E. Radu, and M. Zilhão. “Splitting the Gravitational Atom: Instabilities of Black Holes with Synchronized/Resonant Hair”. *Phys. Rev. D* 113 (30 March 2026). [10.1103/1btp-58hy](https://arxiv.org/abs/10.1103/1btp-58hy)
- J. Ferreira, T. Barreiro, J. Mimoso, and N. J. Nunes. “Testing Λ -free $f(Q)$ cosmology”. *Phys. Rev. D* 108 (6 Sept. 2023). [10.1103/PhysRevD.108.063521](https://arxiv.org/abs/10.1103/PhysRevD.108.063521)
- J. Ferreira, T. Barreiro, J. Mimoso, and N. J. Nunes. “Forecasting $F(Q)$ cosmology with Λ CDM background using standard sirens”. *Phys. Rev. D* 105 (12 June 2022). [10.1103/PhysRevD.105.123531](https://arxiv.org/abs/10.1103/PhysRevD.105.123531)

COMMUNICATIONS

TALKS

Dynamics of Black Holes with Resonant Hair XVIII Black Holes Workshop, Lisbon, Portugal	18-19/12/2025 pptx
Black Holes as Particle Accelerators Annual Meeting of PhD Students from Maths Department, University of Aveiro, Portugal	24/10/2025 pptx
(In)stability of Q-Hairy Black Holes NewFunFICO Meeting, Valência, Spain	20-22/5/2025 pptx
Non-linear Superradiance of Charged Black Holes: Beyond Spherical Symmetry VI Amazonian Symposium on Physics, Belém, Brazil	18-22/11/2024 pptx , pdf

Using Standard Sirens to test non-metric theories of gravity

XVI Black Holes Workshop, Porto, Portugal

19-20/12/2023

[pptx](#), [pdf](#)

Forecasting $F(Q)$ Cosmology with Λ CDM Background using Standard Sirens

Iberian Cosmology Meeting, Barcelona, Spain

4-6/5/2022

[recording](#), [pdf](#), [pptx](#)

SEMINARS

Fate of Black Holes with Resonant Hair

Faculty of Sciences of the University of Lisbon, Portugal

12/12/2025

[pptx](#)

Cosmological Tests on Non-Metricity Based Theories of Gravity

University of Aveiro, Portugal

20/09/2023.

[recording](#), [pptx](#), [pdf](#)

POSTERS

Splitting the (Gravitational) Atom

New Frontiers in Numerical Relativity, Mallorca, Spain

21-25/07/2025

[pdf](#)

Testing Λ -Free $f(Q)$ Cosmology

Dark Matter and Stars, Lisbon, Portugal

3-5/05/2023

[pdf](#)

>_ DIGITAL SKILLS

High Level Languages Julia, Python, Mathematica

Multimedia Editing GIMP, Photoshop, Inkscape, Kdenlive, Audacity

Low Level Languages C, Fortran, CUDA

Web Development Quarto, Hugo

I am an avid Linux user, with experience using it on personal machines, servers, clusters and microcontrollers. I work mostly on a terminal based ecosystem, creating and making use of terminal utilities to develop my projects.

👤 ACADEMIC ACTIVITIES

Supervisor at the IAstro Summer Internships 2023

Institute of Astrophysics and Space Sciences

07/2023

Supervisor of two undergraduate projects that consisted on constraining modified gravity cosmological models with current and forecast datasets coming from different sources.

Podcast “Physically Speaking”

12/2020 - 12/2022

[Fisicamente Falando](#) [Fisicamente Falando](#) [fisicamentefalando.nfep](#) [fisicamente_falando_](#)

A student driven podcast created during the pandemic where we interviewed faculty members to speak about their lives, both inside and outside of academia. I am in charge of the audio and video editing and also take part in some of the interviews.

Member of the Student Body for Physicists and Physical Engineers

10/2020 - 10/2021

[nfef-fcul.com](#) [NFEFFCUL](#) [nfeffcul](#)

I was in charge for the development of the digital tools of the student body of physicists and physical engineers. This included the website, the online repository (which is made up of student provided content) and the Discord channel.

📡 OUTREACH

Pint of Science 2026

20/05/2026

[Presentation \(.pptx\)](#)

Talk at the Pint of Science 2026 in Aveiro, Portugal.

European Researchers' Night 2025

26/09/2025

Coordinator of the participation of Gr@v in the European Researcher's Night in Aveiro, Portugal. Using digital content, we explained the physics behind Gravitational Waves and lensing around compact objects.

“Echoes and Shadows: How to Detect Black Holes”

03/05/2025

[Presentation \(.pptx\)](#)

A Presentation, for high school students, on what are black holes and how to find them.